

Technical Supplement S1 - PFAS Method Validation

MV-PFAS-24-017 Rev A | LC-MS/MS validation of 12 PFAS in finished drinking water

LAB	METHOD BASIS	INSTRUMENT	STUDY DATES
North River Environmental Laboratory	EPA 533 modified	SCIEX 6500+ QTRAP	2024-04-08 to 2024-04-19

SCOPE AND METHOD CONDITIONS

Isotope dilution LC-MS/MS, ESI negative. Column: Waters Acquity BEH C18, 2.1 x 50 mm, 1.7 um. Injection: 5 uL.
Mobile A: 5 mM ammonium acetate in water. Mobile B: methanol.

WORKFLOW



Time min	%B	Flow mL/min	SampleID	Matrix	Collected	Received	pH	Cl mg/L	Vol mL	Condition
0.0	35	0.30	FB-240408-01	Field blank	04-08 09:10	04-08 15:42	6.8	<0.02	252	OK
0.5	35	0.30	DW-240408-02	Finished water	04-08 09:25	04-08 15:42	7.2	<0.02	251	OK
6.0	95	0.35	DW-240408-03	Finished water	04-08 09:40	04-08 15:42	7.1	<0.02	249	OK
8.5	95	0.35	DW-240409-04	Finished water	04-09 10:05	04-09 16:20	7.4	<0.02	250	OK
8.6	35	0.30	DW-240409-05	Finished water	04-09 10:20	04-09 16:20	7.3	<0.02	250	OK
11.0	35	0.30	LRB-240410-01	Lab reagent blank	04-10 07:30	04-10 07:30	6.9	<0.02	250	Prepared
			LCS-240410-01	Lab control sample	04-10 07:35	04-10 07:35	7.0	<0.02	250	Prepared
			MSD-240410-03	Matrix spike dup.	04-10 07:40	04-10 07:40	7.1	<0.02	250	Prepared

Calibration Model and Transition Table

Equations, transitions, calibration range, LOD, and LOQ

R = A_{native} / A_{IS}

C_{sample} = ((R - b) / m) * DF

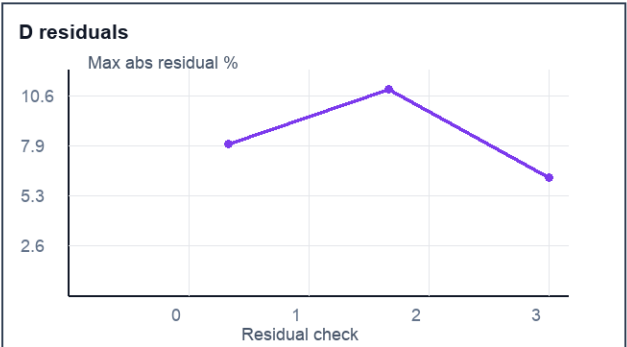
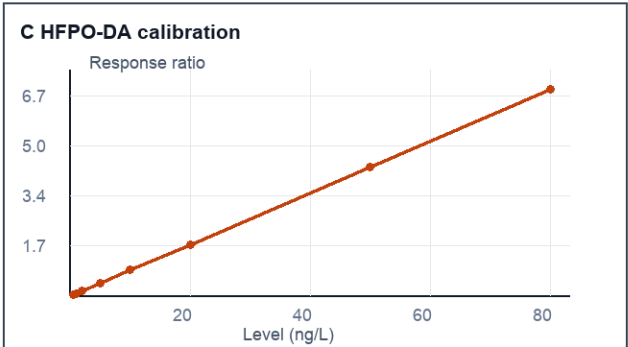
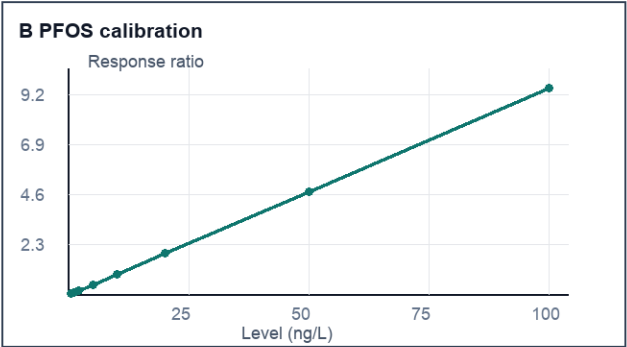
LOD = t_(n-1,0.99) * SD_{low}

U95 = k * sqrt(u_{cal}² + u_{rep}² + u_{vol}² + u_{rec}²), k=2

Analyte	IS	Quant.	Qual.	RT	m	b	R2	Range	LOD	LOQ
PFBA	13C4-PFBA	213>169	213>119	2.18	0.0921	0.0030	0.9987	0.50-80	0.16	0.50
PFPeA	13C5-PFPeA	263>219	263>169	2.74	0.0875	0.0028	0.9991	0.50-80	0.13	0.50
PFHxA	13C5-PFHxA	313>269	313>119	3.32	0.0814	0.0041	0.9985	0.50-80	0.18	0.50
PFHpA	13C4-PFHpA	363>319	363>169	3.86	0.0778	0.0037	0.9990	0.50-80	0.15	0.50
PFOA	13C8-PFOA	413>369	413>169	4.41	0.0749	0.0044	0.9994	0.25-80	0.08	0.25
PFNA	13C9-PFNA	463>419	463>169	4.95	0.0712	0.0051	0.9989	0.25-80	0.09	0.25
PFDA	13C6-PFDA	513>469	513>219	5.48	0.0686	0.0048	0.9986	0.25-80	0.10	0.25
PFBS	18O2-PFBS	299>80	299>99	3.05	0.1033	0.0062	0.9992	0.50-100	0.17	0.50
PFHxS	18O2-PFHxS	399>80	399>99	4.63	0.0968	0.0075	0.9988	0.50-100	0.20	0.50
PFOS	13C8-PFOS	499>80	499>99	5.64	0.0915	0.0081	0.9982	0.50-100	0.22	0.50
6:2 FTS	13C2-6:2FTS	427>407	427>81	5.19	0.0642	0.0057	0.9979	1.00-100	0.31	1.00
HFPO-DA	13C3-HFPO-DA	329>285	329>169	3.71	0.0839	0.0035	0.9993	0.50-80	0.14	0.50

Calibration Curves and Residuals

Four-panel calibration page with residual table



Analyte	Max abs residual %	Failing points	Reviewer note
PFOA	7.8	0	Weighted 1/x2 retained
PFOS	10.6	0	High calibrator within +/-15%
HFPO-DA	6.1	0	No curvature observed

Table S4. Accuracy and Precision - Part 1

Fortified reagent-water recovery and precision; continued on page 5

Analyte	Low Rec	Low RSD	Mid Rec	Mid RSD	High Rec	High RSD	N	Status
PFBA	96	8.4	101	4.6	99	3.8	7	Pass
PFPeA	93	9.1	98	5.2	101	4.1	7	Pass
PFHxA	91	10.3	97	5.8	100	4.3	7	Pass
PFHpA	95	7.9	99	4.9	102	4.0	7	Pass
PFOA	98	6.8	103	3.7	101	3.2	7	Pass
PFNA	97	7.2	102	4.1	100	3.6	7	Pass
PFDA	92	11.5	96	6.3	99	4.9	7	Pass

Footnotes

Acceptance: 70-130% recovery and RSD <= 20% unless method-specified criterion is tighter. Low level equals 2x LOQ for PFOA/PFNA/PFDA and equals LOQ for remaining analytes.

Continuation

Rows 8-12 continue on page 5 under the same Table S4 header. The header is repeated there for readability.

Table S4 Continued + Matrix Spike Recovery

Continuation rows, matrix spike duplicate, RPD, and qualifier flags

Analyte	Low Rec	Low RSD	Mid Rec	Mid RSD	High Rec	High RSD	N	Status
PFBS	104	9.7	99	4.8	98	4.0	7	Pass
PFHxS	101	8.9	100	4.4	97	3.9	7	Pass
PFOS	89	12.6	95	6.7	96	5.1	7	Pass
6:2 FTS	86	14.8	92	7.4	94	6.3	7	Pass
HFPO-DA	99	7.5	104	4.2	103	3.5	7	Pass

MATRIX SPIKE / DUPLICATE TABLE

SampleID	Analyte	Native	Spike	MS Rec	MSD Rec	RPD	Flag
DW-240408-03	PFOA	3.42	10.00	96	98	2.1	None
DW-240408-03	PFOS	0.74	10.00	91	88	3.4	J-native below 2x LOQ
DW-240408-03	PFHxS	1.16	10.00	102	99	3.0	None
DW-240408-03	HFPO-DA	<0.50	10.00	105	106	1.0	U-native
DW-240408-03	6:2 FTS	<1.00	10.00	84	87	3.5	Low bias monitored

Batch Sequence, QC State, and Review Trail

Sequence table with reinjection state and review signoff

Seq	Type	Vial	SampleID	Time	IS OK	Carryover	State
1	Solvent blank	A01	SBLK-01	08:10	Yes	Yes	Accepted
2	Calibration	A02	CAL-0.5	08:23	Yes	Yes	Accepted
3	Calibration	A03	CAL-1	08:36	Yes	Yes	Accepted
4	Calibration	A04	CAL-2	08:49	Yes	Yes	Accepted
5	Calibration	A05	CAL-5	09:02	Yes	Yes	Accepted
6	Calibration	A06	CAL-10	09:15	Yes	Yes	Accepted
7	Calibration	A07	CAL-20	09:28	Yes	Yes	Accepted
8	Calibration	A08	CAL-50	09:41	Yes	Yes	Accepted
9	Calibration	A09	CAL-80	09:54	Yes	Yes	Accepted
10	Blank	B01	LRB-240410-01	10:07	Yes	Yes	Accepted
11	LCS	B02	LCS-240410-01	10:20	Yes	Yes	Accepted
12	Sample	B03	FB-240408-01	10:33	Yes	Yes	Accepted
13	Sample	B04	DW-240408-02	10:46	Yes	Yes	Accepted
14	Sample	B05	DW-240408-03	10:59	Yes	Yes	Accepted
15	MS	B06	DW-240408-03-MS	11:12	Yes	Yes	Accepted
16	MSD	B07	DW-240408-03-MSD	11:25	Yes	Yes	Accepted
17	CCV	B08	CCV-10	11:38	Yes	Yes	Accepted
18	Sample	B09	DW-240409-04	11:51	No	Yes	Reinject
19	Sample	C01	DW-240409-04-RI	12:18	Yes	Yes	Accepted
20	Sample	C02	DW-240409-05	12:31	Yes	Yes	Accepted

Review trail

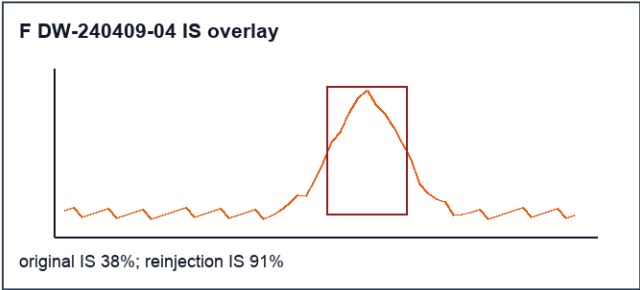
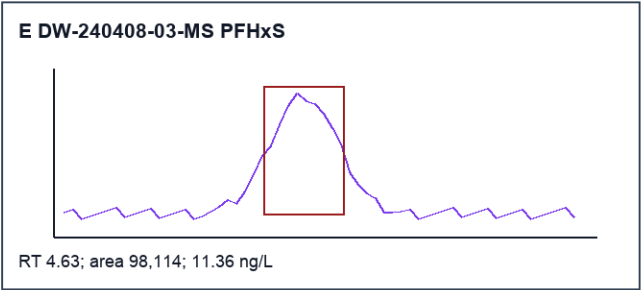
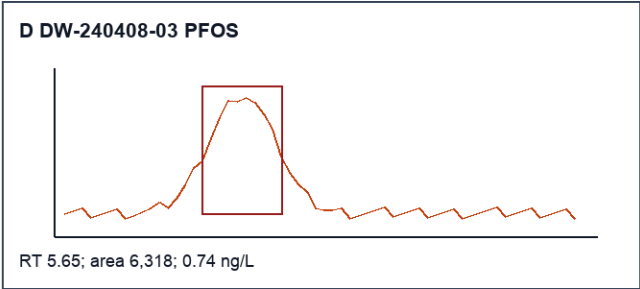
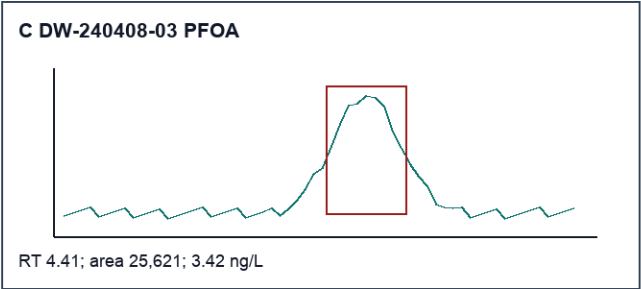
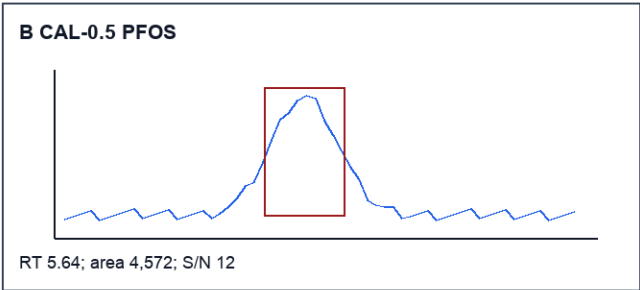
Initial analyst review: M. Iyer, 2024-04-19 14:05. Technical review: L. Chen, 2024-04-22 09:40. QA release: R. Patel, 2024-04-22 16:10.

Reinjection state

Seq 18 DW-240409-04 failed internal standard recovery and is marked Reinject. Seq 19 DW-240409-04-RI is the accepted reinjection used for final results.

Chromatogram Plate S7

Six-panel chromatogram-style visual evidence with retention windows



Analyte	RT min	Window
PFOA	4.41	+/-0.08
PFHxS	4.63	+/-0.08
6:2 FTS	5.19	+/-0.10
PFOS	5.64	+/-0.10

Final Results, Uncertainty, and Cross-References

Regulatory result table, qualifiers, uncertainty budget, and references

SampleID	Analyte	Result	Qual.	U95	Reference
FB-240408-01	PFOA	<0.25	U		Table S2
FB-240408-01	PFOS	<0.50	U		Table S2
DW-240408-02	PFOA	2.18		0.42	Fig S7C
DW-240408-02	PFOS	0.62	J	0.18	Fig S7D
DW-240408-02	PFHxS	0.91	J	0.21	Table S5
DW-240408-03	PFOA	3.42		0.55	Fig S7C
DW-240408-03	PFOS	0.74	J	0.19	Fig S7D
DW-240408-03	HFPO-DA	<0.50	U		Table S2
DW-240409-04	PFOA	1.06	J	0.25	Seq 19
DW-240409-04	PFOS	<0.50	U		Seq 19
DW-240409-05	PFOA	4.87		0.72	Table S8
DW-240409-05	PFOS	1.34		0.29	Table S8

Component	PFOA %	PFOS %	HFPO-DA %
Calibration	5.2	7.4	4.8
Repeatability	4.1	6.9	3.7
Volume	1.5	1.5	1.5
Recovery	3.6	5.1	3.2
Combined u	7.7	11.4	6.9
Expanded U k=2	15.4	22.8	13.8

Final-state links

DW-240409-04 final results reference Seq 19, not Seq 18. U means not detected, J means estimated, RI means reinjection, MS/MSD indicate matrix spike and duplicate.